

FILIPPO MARCANTONI

Robotics Software Engineer | Computer Vision & 3D Perception | Industrial Automation | Robot Learning | Physical AI
filippo.marcantoni@gmail.com | (508) 365-9842 | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

SUMMARY

Robotics engineer with an M.S. in Robotics Engineering from Worcester Polytechnic Institute and hands-on R&D experience across robotics software, autonomous navigation, computer vision, robot learning, embedded robotics, industrial automation, teleoperation, and swarm robotics. Experienced in building and integrating systems across perception, planning, learning, simulation, sensing, communication, and real-world testing. Focused on practical robotic systems that are reliable, adaptable, and effective in dynamic and previously unseen environments.

EDUCATION

Worcester Polytechnic Institute (WPI)

M.S. Robotics Engineering | GPA 3.90/4.00

B.S. Robotics Engineering | GPA 3.89/4.00, Summa Cum Laude

Worcester, MA, USA

Aug 2024 – May 2026

Aug 2021 – May 2025

EXPERIENCE

AI & Industrial Robotics R&D Intern | B&R Industrial Automation, Austria

Jun 2025 – Aug 2025

- Developed an LLM-based code-generation copilot for B&R Automation Studio using Transformers, RAG, Azure AI Search, and domain-specific Motion Structured Text (Motion ST) examples to support industrial robotics and automation workflows.
- Designed the data-ingestion and retrieval backend using domain-aware metadata, custom syntax-aware chunking for Structured Text and Motion ST, and hybrid dense and sparse retrieval with PostgreSQL, pgvector, and BM25.

Machine Vision & AI R&D Intern | Makro Labelling SRL, Italy

May 2024 – Aug 2024

- Developed C++ and Python machine-vision pipelines for bottle orientation, label alignment, and defect inspection on high-speed industrial labeling lines.
- Benchmarked classical and deep-learning vision models across bottle formats, lighting conditions, and label shifts to identify failure modes, validate robustness under production constraints, and improve automated quality-control performance.

Automation Software Engineer Intern | Makro Labelling SRL, Italy

May 2023 – Aug 2023

- Supported commissioning, testing, and troubleshooting of PLC-controlled labeling machines using B&R Automation Studio, Structured Text, Ladder Logic, and Visual Components.
- Performed alarm validation, sensor calibration, and activation testing to verify control sequences, sensor signals, and machine readiness.

RESEARCH

PiELO: Reactive Swarm Programming Language | NEST Lab, WPI

Aug 2024 – May 2025

- Co-developed PiELO, a reactive C++ programming language for robot swarms, implementing core compiler and runtime components including a tokenizer, parser, AST-to-bytecode compiler, stack-based virtual machine, reactive closures, and mark-and-sweep garbage collection.
- Implemented distributed coordination mechanisms, including a last-writer-wins shared dictionary, and validated consensus, barrier, and leader-election behaviors in ARGoS3 and on Khepera IV robots with up to 90% packet loss.

VR-Based Multimodal Teleoperation Interface | HiRo Lab, WPI

Jan 2025 – May 2025

- Developed a human-in-the-loop VR teleoperation interface for a robotic nursing assistant using Unity, ROS, Meta Quest head-motion mirroring, real-time multi-camera streaming, Whisper voice commands, and ZED-Mini 3D object detection.
- Integrated viewpoint control, speech-based camera switching, and perception-driven camera selection to improve operator situational awareness during remote robotic manipulation.

SELECTED PROJECTS

Robot Learning: Reinforcement & Imitation Learning

- Implemented policy iteration, REINFORCE, A2C, and A3C for simulated control and robotic manipulation tasks, including LunarLander and PyBullet Kuka vision-based grasping, reaching 55% grasp success with a CNN actor-critic policy.
- Trained imitation-learning policies in robomimic on 60 PickPlaceCan demonstrations, improving success from 0.02 with baseline behavioral cloning to 0.80 with LSTM behavioral cloning and 0.90 with diffusion policy.

Classical MSCKF & Deep Visual-Inertial Odometry

- Implemented a stereo MSCKF visual-inertial odometry backend with IMU propagation, camera-state cloning, stereo feature tracking, and nullspace EKF updates, evaluated on EuRoC and RPG datasets.
- Trained IMU-only, vision-only, and fused deep VIO networks on synthetic Blender trajectories, evaluating sensor-fusion behavior with geodesic SO(3) and trajectory-composition losses.

TECHNICAL SKILLS

Languages: Python, C++, C, C#, MATLAB, Java, JavaScript, Buzz.

Robotics, Embedded & Simulation: ROS/ROS2, Gazebo, RViz, MuJoCo, PyBullet, ARGoS3, ESP32, Arduino, Raspberry Pi, MSP430, OpenMV, Unity, Vicon, VR/XR, Automation Studio.

ML & AI: PyTorch, TensorFlow, Scikit-learn, CUDA, Transformers, LLMs, RAG, DL, Reinforcement Learning, Imitation Learning.

Computer Vision & 3D Perception: OpenCV, SLAM, Visual-Inertial Odometry, MSCKF, SfM, NeRF, RANSAC, PnP, Bundle Adjustment, Homography, Object Detection, Segmentation, Monocular Depth, Pose Estimation, Optical Flow, Point Clouds.

Tools & Infrastructure: Linux, Git, Docker, Jupyter, Azure AI, AWS, Simulink, React, Node.js, PostgreSQL, MySQL, HPC/Slurm.

CAD & Design: SolidWorks, Fusion 360.